

WILLIAM ALEXAKIS

Athens, Greece

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EDUCATION

St. Catherine's British School

Athens, Greece

International Baccalaureate Diploma Programme

September 2026 — June 2028

Computer Science HL, Mathematics Analysis & Approaches HL, Physics HL

IGCSE Programme

September 2024 — June 2026

Computer Science, Mathematics Extended, Physics Triple Award

PROJECTS

self-attention.c

June 2026

github.com/williamalexakis/self-attention.c

- Wrote a scaled dot-product self-attention implementation in C with zero dependencies in ~100 LOC, featuring Q/K dot product computation, score normalization by root of d_k to prevent softmax saturation, softmax weight distribution, and weighted value aggregation.

Phase Programming Language

August 2025 — December 2025

github.com/williamalexakis/phase

- Wrote a statically-typed programming language in C with zero dependencies in ~4,800 LOC, featuring a 25-opcode stack-based virtual machine, 5 primitive types, and tuple destructuring.
- Built a full compilation pipeline including a lexer, recursive-descent parser, static type checker, bytecode generator, and a stack-based virtual machine with scoped call frames.
- Implemented 23 error types with source-mapped diagnostics displaying exact file/line/column, visual markers, and diff-style fix suggestions.

EXTRACURRICULAR ACTIVITIES

Open-Source Software Contributions

June 2026 — Present

Contributor

- Implemented syntax error handler with source references for 6 error types in C++ for a programming language, reducing redundant code and improving error UX (github.com/omnimistic/algo-lang/pull/13).

Initiative for Revision of AI Usage Policy

March 2026 — April 2026

Founder & Organizer

- Led an initiative to revise the school's firewall on generative AI, citing IBO anti-restrictive policy to advocate for ~200 IB students.
- Acquired ~140 student signatures and 6 staff endorsements including the Head of Maths Department and Deputy Head of Upper School through physical cold approaches.
- Negotiated directly with the Senior Leadership Team via the Head of Digital Innovations, resulting in a policy review for expanded AI access in the 2026/2027 academic year.

StCatsHacks Line Robot Competition

December 2023 — December 2025

Team Leader & Lead Programmer

- Won 1st place for 3 consecutive years (2023 — 2025) leading a 3-person team.
- Designed a 3-sensor line-following algorithm to resolve simultaneous sensor activation on reflex-angle turns, outpacing the standard PID approach used by competing teams on both speed and accuracy.

ADDITIONAL INFORMATION

Technical Skills

C, Python, Git

Spoken Languages

Fluent English, Native Greek